

FACULTY OF SCIENCE, ENGINEERING AND COMPUTING

School of Computer Science & Mathematics

***BSc DEGREE
IN***

***Computer science in network and network
security***

PROJECT INTERIM REPORT

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Project Title: A Secure Hybrid network Real-Time Data Streaming and Storage
System for TV/Multimedia Broadcasting/Videography and Audio.

Project Type: Build

Date: 2026/07/20

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Did you discuss and agree the viability of your project idea with your supervisor?

Yes

Did you submit a draft of your proposal to your supervisor?

Yes

Did you receive feedback from your supervisor on any submitted draft?

Yes

Abstract

This interim report presents the design of a low-cost hybrid-network prototype for live video broadcasting and cloud recording. The proposed system captures a webcam feed, encodes it for live delivery, records it in short video segments and uploads completed segments to cloud storage. Two ordinary Internet connections are used to represent a 5G link and a satellite link. A monitoring component observes network performance and selects a backup link when the active link remains below a configurable bandwidth threshold. The project is intentionally scoped as link failover rather than commercial multi-link bonding. At the interim stage, the requirements, feasibility study, system architecture, implementation plan and evaluation plan have been completed. The remaining work is to configure the streaming pipeline, automate cloud uploads, implement and test failover, and evaluate continuity, switching delay and upload reliability. The intended result is an understandable, reproducible educational prototype that demonstrates resilient live streaming without requiring specialist telecommunications equipment.

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Proposed Figure Secure Hybrid network Real-Time Data Streaming and Storage System

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Table 3

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Glossary of Terms

1. Introduction & Literature Review

1.1 Introduction

1.2 Background and Motivation

I've been working in the television broadcast engineering field as a for almost a decade. One of the challenging tasks in the practical exercises is broadcasting and recording outdoors in rural or remote locations and special events.

Although the system that we and most organisations currently use to send data (video embedded with audio) has no cloud storage. It only capable of live streaming.

The project will show the principles of hybrid connectivity, cloud uploading, live streaming and authentication. The project uses two links. A phone hotspot or Wi-Fi service representing the 5G, while an ethernet, a second hotspot or another Wi-Fi network representing the satellite connectivity. The system rely on those two connections and switches between them depending on the signal strength or the bandwidth.

Covering domains

- *Network Security*
- *Cloud Security*
- *Real-Time Data communication Systems*
- *Data encryption*
- *User authentication systems*

1.3 Problem in brief

One of the downsides the current system is that in order to capture the video, the sender and the receiver required to be in uninterrupted realtime communication and in alert. So it requires another system for uninterrupted coordination (most off the time on cellular network for calls). If by any mean the receiver had failed to capture the video or missed the command from the senders end , there is no way recovering the lost data since it does not provide cloud storage, nor redundancy. It could be during a live stream during an exciting moment at a sports event, it could be an important speech at a political stage or a once a lifetime moment at a wild park. Where importance of the flow of data is high.

1.4 Aim & Objectives

1.4.1 Aim

The aim of this project is to design and implement a simplified hybrid 5G and satellite simulated system that captures video, provides live streaming, uploads video to cloud storage and selects a between networks when the data bandwidth is below a selected threshold.

1.4.2 Objectives

- *Develop video capture and encoding system using Python/ Java program.*
- *Automatically upload the capturing video segments to a cloud storage.*
- *Monitor two available network links and switch between routes when the active connection is below a configured threshold..*
- *Configuring a streaming server to receive the live feed and develop a playback*
- *Making a cloud upload method which Splits recordings into short video files then automatically uploads completed files to a cloud storage.*

1.5 Scope

The project uses a webcam source as the video input, two Internet links representing 5G and satellite connection to switch between depending on the connection bandwidth. Video quality with 720p for a low bitrate suitable for the project.

HTTP/HLS Live Streaming is used for playback, because it transfers an ongoing multimedia session as segments. It allows the network to send the data seamlessly and allows to live stream to start without waiting for the whole session to be complete. Also it helps to reduce data traffic and redundancy since it sends small segments instead of sending a huge bulk of data.

A buffer time will be introduced (around 5ms) so it gives time for the processing time and to resend segments if any is missing.

1.6 Deliverables

A Secure Hybrid network Real-Time Data Streaming and Storage System that is capable of live streaming and cloud uploading in real time. The system will be connected to two networks where switching between them will be dependent on the bandwidth either connections.

Test and result documentation will be included. It contains test plan and test cases, screenshots, logs, and results along with video evidence.

Java/Python scripts used for monitoring, switching and uploading.

Dashboard will include active network connection status, Bandwidth and latency status and live streaming conditions for playback.

An automated cloud upload component. Split the captured video into segments while maintaining a log for upload success or failure details.

1.7 Literature Review

Live video streaming and broadcasting requires the continuous capture, encoding, uninterrupted transmission and delivery of video to viewers/receivers. A major challenge is that live video quality heavily depends on the availability and the quality network connection used. When facing low bandwidth, the stream may buffer or disconnect due to high latency. This is especially important for TV broadcasting, live streamers,, journalists, event organisers and so on. It is an advantage relying on more than one stable Internet connection.

Commercial systems that is broadly used in Sri Lanka such as LiveU LU800 provide seamless service is a benchmark for same level systems. LiveU system bonds multiple cellular, connections to create a more strong uplink.. These systems demonstrate the value of multiple connections for live broadcasting and live streaming. Their main tactical is traffic can use multiple links, which reduces dependence on a single mobile network. Although these systems does not upload to a cloud or contain redundancy. Also continuous coordination between two ends is requited for video capturing and for a successful session. Therefor the proposed system contains these missing advantages from the current system.

At the streaming layer, RTMP, HLS were considered as in RTMP acts as an ingest protocol because it enables the field application to send a continuous encoded video feed to a streaming server. HLS will be used for viewer delivery because it uses a playlist containing sequential media segments, allowing a browser to play an ongoing stream. Its advantage is broad compatibility with web delivery. For this project, reliability and browser access are important. Therefore, HLS is selected for playback.FFmpeg was selected as the main media-processing technology because it supports encoding, RTMP output and HLS/segment generation.

There for in the final design webcam feed will be encoded by FFmpeg, sent to a streaming server using RTMP, delivered to viewers through HLS, recorded in short segments, uploaded to cloud storage and controlled by Python based monitoring.

2. Analysis

The analysis recognised that a single network live streaming/broadcasting setup low bandwidth and high latency. It also identified the risk of losing video data when it only streams video without any redundancy concerns and has no cloud uploading method.

The proposed solution uses two Internet connections representing a 5G and a satellite link. The system monitors the active connection and switches to the secondary connection when the active link reach below a defined threshold. At the same time, it splits the video data into segments and uploads those segments to cloud storage.

*Table 01.
Requirements of the proposed system*

Requirement	Priority	Use
<i>Capture video from an attached webcam.</i>	<i>Must</i>	<i>A preview and encoded video feed are produced.</i>
<i>Send live video to the streaming server.</i>	<i>Must</i>	<i>A viewer can open the live channel in a browser.</i>
<i>Create short recordings and upload them automatically.</i>	<i>Must</i>	<i>Completed segments appear in cloud storage.</i>
<i>Monitor both simulated links.</i>	<i>Must</i>	<i>Measurements and active-link state are logged.</i>
<i>Fail over below a defined threshold.</i>	<i>Must</i>	<i>The preferred link changes after sustained degradation.</i>

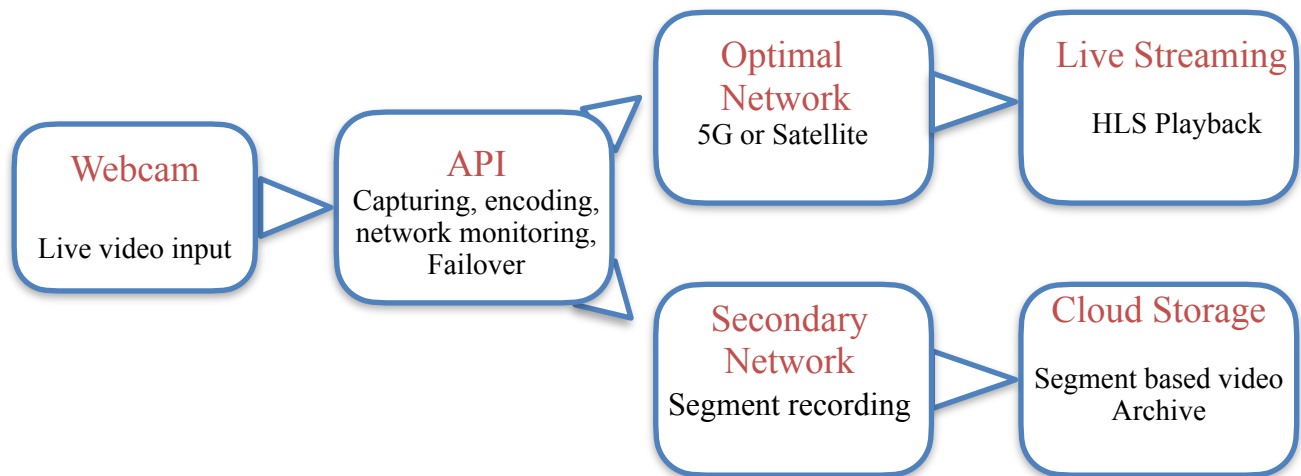
3. Design

3.1 Design Techniques

The design has each segments where each carry responsibilities such as capture and encoding, streaming, segmentation, cloud upload, link measurement, decision making and status display.

Figure 01.

Proposed Figure Secure Hybrid network Real-Time Data Streaming and Storage System



3.2 System Overview

The switching algorithm has a simple rule, measure the quality and the bandwidth of the stream. If the active connection goes below the threshold and the backup score is acceptable, change the network route and restart/continue the stream with the reliable path.

The dashboard will show the active link, latest measurements, threshold, streaming status, latest cloud upload and failover count. This dashboard is not a full network management type tool. it simply provides details and data for demonstration.

Table 02.

Protocols and systems used to achieve the project.

Protocol / system	Use
HTTP	Transfers the HLS playlist and video segments from the streaming server to viewers.
HLS	Delivers the live stream from the server to viewers in a web browser. It uses short video segments.
HTTPS	Used for the dashboard and cloud uploads.
ICMP/ Ping	Checks the network connection's reachable and measures latency so the monitoring script can use it.
IPv4/ IPv6	The operating system uses these routing rules to choose which network connection is the best path
Wifi and Ethernet	One local connection simulating the 5G connection and one wired connection simulating satellite connection.
RTMP	carries the live video feed from the sender to the streaming platform in real time.
Python	Acts as the main controller of the system for decision making and coordinating.
MediaMTX + FFmpeg	For capturing, converting, compressing, recording and for routing.

4. Product Implementation

Technical description of implementation: use of libraries and interfaces; functions used; application of coding principles; critical discussion of coding issues; sophistication of code; code structure; choice of methodologies; modularity

5. Validation

Strategies and outcomes: Clear test strategy; testing framework; evaluation analysis; usability testing; performance testing; black box / white box; functionality; feedback from client

6. Critical Review & Conclusion

6.1 Closing executive summary

The interim review shows that the proposed project is feasible if it remains a failover prototype. The design uses accessible tools while still addressing a meaningful networking problem: maintaining video service and preserving recordings when a preferred link degrades. The most important scope decision is to avoid promising commercial-grade bonding or seamless handover. This keeps the evaluation credible and gives the project a well-defined contribution.

6.2 Conclusion

This interim report presented the proposed design for a hybrid 5G–satellite simulated system for live video streaming and cloud recording. The project addresses the problem of unreliable live broadcasting when a device depends on one Internet connection. The proposed solution uses two ordinary Internet connections to represent 5G and satellite links, monitors their performance, and changes to a backup connection when the active link remains below a defined threshold.

The project aim is to develop a simplified system that captures webcam video, streams it live, records it in segments, uploads recordings to cloud storage and performs automatic network failover. At the interim stage, the aim has been achieved at the planning and design level. The project problem has been defined, relevant technologies have been reviewed, requirements have been identified, the system architecture has been designed, and an implementation and validation plan has been produced.

Future work could improve the system by implementing Multipath TCP or other multi-path technologies, adding adaptive bitrate streaming, using Secure Reliable Transport, improving dashboard features, encrypting recordings end-to-end and testing with real 5G or satellite Internet services. A mobile application and support for multiple cameras could also be investigated in future.

References / Bibliography

Here you should give details of citations that you have used in the text. An entry in list of reference generally includes information such as Author, Year, Title of the Article, Name of Journal/conference, page numbers. There are various reference and citation styles, but you should use the one shown in the sample.

Appendices

Appendix A – Proposed System Requirements

Hardware requirements

- *Laptop or desktop computer*
- *Built-in or USB webcam*
- *Two Internet connections:*
 - *Mobile hotspot/Wi-Fi to simulate 5G*
 - *Ethernet, second Wi-Fi, or second hotspot to simulate satellite*
- *Optional external microphone*

Software requirements

- *Python*
- *FFmpeg*
- *Nginx with RTMP support*
- *Flask dashboard*
- *AWS account and S3 bucket*
- *Web browser*

Table 3
Appendix B

<i>Task</i>	<i>Status at interim stage</i>
<i>Define project problem and objectives</i>	<i>Completed</i>
<i>Literature review</i>	<i>Completed</i>
<i>Requirement analysis</i>	<i>Completed</i>
<i>System architecture design</i>	<i>Completed</i>
<i>Set up Python and development environment</i>	<i>Planned</i>
<i>Implement webcam capture</i>	<i>Planned</i>
<i>Configure FFmpeg and streaming server</i>	<i>Planned</i>
<i>Implement cloud upload</i>	<i>Planned</i>
<i>Implement network monitoring</i>	<i>Planned</i>
<i>Implement automatic failover</i>	<i>Planned</i>
<i>Testing and evaluation</i>	<i>Planned</i>
<i>Final report and presentation</i>	<i>Planned</i>